Comprehensive Analytical Report: Uttar Pradesh State E-Waste Management and Recycling Scorecard (2025)

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Abstract

This report presents a comprehensive analytical assessment of e-waste generation, collection efficiency, recycling infrastructure, regulatory compliance, environmental safeguards, and public awareness across Uttar Pradesh under the E-Waste (Management) Rules, 2022. Based on the official 100-point assessment framework, Uttar Pradesh achieves an overall score of 20.5/100, placing the state in the Critical performance category.

Prepared with reference to: Uttar Pradesh Pollution Control Board (UPPCB) and Central Pollution Control Board (CPCB)

Regulatory Framework: *E-Waste (Management) Rules, 2022* (Schedule I & CPCB Guidelines)

Dataset: `State_EWaste_Tracker_Final.xlsx`

Project Portal: https://project.seucra.tech

I. Abstract

This report presents a comprehensive analytical assessment of e-waste generation, collection efficiency, recycling infrastructure, regulatory compliance, environmental safeguards, and stakeholder awareness across Uttar Pradesh. The evaluation follows the performance framework established under the *E-Waste (Management) Rules, 2022* and relevant Central Pollution Control Board (CPCB) guidelines. Based on the official 100-point assessment framework, Uttar Pradesh achieves an overall score of **20.5/100**, placing the state within the **Critical Performance Band (0–44 points)**. The findings identify substantial deficiencies in inventory management, formal collection systems, recycling infrastructure, and informal sector integration, while highlighting strengths in Extended Producer Responsibility (EPR) compliance. Strategic recommendations are proposed to improve statewide compliance and environmental performance.

II. Index Terms

Electronic Waste, E-Waste Management, Circular Economy, Recycling, Environmental Governance, Extended Producer Responsibility (EPR), Uttar Pradesh, CPCB.

III. I. Introduction

Electronic waste (e-waste) represents one of the fastest-growing waste streams worldwide, driven by rapid technological advancement, increasing consumer demand, and shortened product life cycles. Effective management of e-waste is essential to minimize environmental contamination, recover valuable materials, and promote sustainable resource utilization.

The Government of India regulates e-waste through the **E-Waste (Management) Rules, 2022**, administered by the Central Pollution Control Board (CPCB) and State Pollution Control Boards (SPCBs). This report evaluates Uttar Pradesh's performance using the official state-level assessment framework based on quantitative indicators covering generation, collection, recycling capacity, regulatory compliance, environmental safeguards, and public awareness.

The analysis utilizes the dataset:

Dataset: *State_EWaste_Tracker_Final.xlsx*

Hosting Portal: https://project.seucra.tech

IV. II. State-Level Performance Overview

Table I summarizes the principal statewide statistics.

Table I
Statewide E-Waste Statistics

Parameter	Value
Total Annual E-Waste Generated	**466,578 tonnes**
Total E-Waste Collected	**376,896.74 tonnes**
Collection Rate	**80.78%**
Authorized Recycling Capacity	**388,000 tonnes/year**
Authorized Recycling Units	**125**
Registered EPR Producers	**9,956**
Major Informal Hotspots	**4**
Overall State Score	**20.5 / 100**
Performance Band	**Critical**

The statewide assessment indicates that although collection efficiency remains relatively high, significant deficiencies exist in inventory completeness, formal recycling infrastructure, environmental safeguards, and integration of informal recycling activities.

V. III. Methodology

The assessment follows the performance indicators prescribed under the **E-Waste (Management) Rules, 2022** together with CPCB implementation guidelines.

The evaluation consists of seven weighted performance pillars totaling 100 points.

Table II
Performance Evaluation Framework

Indicator	Maximum Weight	Score
Generation and Inventory	15.0	0.0
Collection Performance	20.0	0.0
Formal Recycling and Treatment	20.0	0.0
Regulatory Compliance	15.0	15.0
Informal Sector Integration	10.0	0.0
Environmental and Health Safeguards	10.0	5.0
Awareness and Outreach	10.0	0.5
Total	**100.0**	**20.5**

VI. IV. Results and Analysis

A. A. Generation and Inventory

The estimated annual e-waste generation for Uttar Pradesh is **466,578 tonnes**.

Population-intensive districts account for the highest projected generation volumes. The five largest contributors include:

1. Prayagraj
2. Moradabad
3. Ghaziabad
4. Lucknow
5. Kanpur Nagar

Although statewide generation estimates are available, district-level digital inventory systems remain incomplete. The absence of standardized reporting for individual waste categories—including Information Technology equipment, consumer electronics, batteries, and lighting products—results in a zero score under the inventory completeness criterion.

B. B. Collection and Recycling Infrastructure

The authorized recycling infrastructure has a combined annual capacity of **388,000 tonnes**, while annual generation is estimated at **466,578 tonnes**.

Consequently, the state exhibits a capacity deficit of approximately:

78,578 tonnes/year

The actual quantity processed during the reporting period reached **376,896.74 tonnes**, corresponding to a collection efficiency of **80.78%**.

Major authorized recycling facilities include:

Organization	Location	Capacity (tonnes/year)	Processed (tonnes)
Attero Recycling Pvt. Ltd.	Noida	120,000	96,000
Sims Recycling Solutions	Noida	35,000	28,000
Greenex Eco Recycling	Ghaziabad	24,000	18,500
Horizon Recycling	Moradabad	18,000	14,200
Bharat Oil & Waste Management	Kanpur	15,000	11,000

Despite relatively high operational throughput, the evaluation framework assigns no score because infrastructure coverage and collection mechanisms do not fully satisfy the prescribed regulatory benchmarks.

C. C. Regulatory Compliance

Regulatory compliance constitutes the strongest performing indicator.

A total of **9,956 producers** are registered on the CPCB National EPR Portal, supported by periodic inspections conducted by the Uttar Pradesh Pollution Control Board.

Accordingly, the state receives the maximum score of **15/15** under this performance category.

D. D. Environmental Safeguards and Informal Sector

The assessment identifies four major informal recycling clusters requiring continued regulatory intervention.

1) 1) Moradabad Circuit Cluster: The Karula–Nawabpura region remains a significant informal printed circuit board processing hub utilizing acid leaching techniques. Environmental monitoring has detected elevated concentrations of lead (Pb), cadmium (Cd), and chromium (Cr) in surrounding soil and water bodies.

2) 2) Loni–Seelampur Border: This corridor remains an active enforcement zone targeting illegal cable burning and unauthorized dismantling operations.

3) 3) Dasna–Masuri Cluster: Multiple dismantling units have received closure notices under the Air Act and Water Act due to non-compliance.

4) 4) Kanpur Informal Market: The Uttar Pradesh Pollution Control Board has submitted a corrective action plan before the National Green Tribunal addressing unauthorized e-waste processing activities.

Overall, environmental safeguards achieve only partial compliance, primarily due to continuing informal recycling operations and associated environmental risks.

VII. V. Discussion

The assessment demonstrates a substantial imbalance between regulatory administration and operational implementation.

While Extended Producer Responsibility registration has achieved near-complete compliance, operational deficiencies persist in:

- district-level inventory digitization,
- formal collection networks,

- recycling capacity expansion,
- informal sector formalization,
- environmental monitoring,
- statewide awareness initiatives.

These deficiencies collectively reduce the overall state score despite relatively high collection volumes.

VIII. VI. Recommendations

The following strategic actions are recommended to improve statewide performance:

1. Implement mandatory digital inventory reporting across all 75 districts using standardized electronic reporting systems.
2. Formalize informal recycling workers in Moradabad and Loni through cooperative collection centers, occupational health programs, and authorized dismantling facilities.
3. Expand authorized recycling infrastructure to eliminate the estimated annual processing deficit of **78,578 tonnes**.
4. Increase statewide public awareness campaigns from a single annual initiative to at least twenty district-level programs each year.
5. Strengthen environmental surveillance around identified informal recycling hotspots through continuous monitoring of soil, groundwater, and air quality.
6. Improve integration between district administrations, municipal bodies, producers, and authorized recyclers to strengthen collection logistics.

IX. VII. Conclusion

Uttar Pradesh currently achieves an overall performance score of **20.5/100**, placing the state within the **Critical** category under the CPCB assessment framework. Although regulatory compliance through the Extended Producer Responsibility mechanism remains robust, major improvements are required in inventory management, recycling infrastructure, environmental protection, and public engagement.

Targeted investments in infrastructure expansion, digital reporting systems, and formalization of the informal recycling sector are expected to substantially improve environmental outcomes while increasing future compliance scores.

X. Acknowledgment

This report was generated using the **State E-Waste Tracker Final Analytics Engine** based on the dataset *State_EWaste_Tracker_Final.xlsx*.

XI. References

[1] Ministry of Environment, Forest and Climate Change, *E-Waste (Management) Rules, 2022.*

[2] Central Pollution Control Board (CPCB), *Guidelines for Environmentally Sound Management of E-Waste.*

[3] Uttar Pradesh Pollution Control Board (UPPCB), *Annual Environmental Compliance Reports.*

[4] State E-Waste Tracker Analytics Dataset, *State_EWaste_Tracker_Final.xlsx.*